

Task 9: File Input Output

Instead of just printing the final report card text to your screen using `System.out`, we will simultaneously write that exact text data to a permanent file named **report_cards.txt**. Each time you add a new student, the program will **append** (add onto) the bottom of that file so your history is never lost.

Note:

No changes to class `Student` and class `GraduateStudent` for adding File Input Output. Only adjust main method to automatically save these report cards to your hard drive, we will introduce **File I/O** (Input/Output) using Java's built-in `FileWriter` and `IOException` error handling.

What We Are Adding in This Final Step:

1. **New Package Imports:** We add `import java.io.FileWriter;` and `import java.io.IOException;` to the very top of our file to pull Java's file-writing tools off the digital library shelf.
2. **The FileWriter Tool:** We open a writer channel inside our report generation segment. By passing `true` as a configuration parameter (`new FileWriter("report_cards.txt", true)`), we tell Java to update and append to the text file rather than wiping it clean every loop cycle.
3. **Mandatory File Try-Catch:** Writing files to a hard drive carries real-world risks (like your hard drive running out of space or being locked). Java forces us to wrap our file-writing code inside a try-catch block to handle an `IOException` safely without crashing.

Verifying Your Work

Once you compile and run this program, add a couple of student records, and select option 3 to exit.

Look inside your project's main folder—you will see a brand-new file named `report_cards.txt` created entirely by your code. If you open it with Notepad or any basic text editor, you will find your clean academic records saved exactly as specified on your disk [local].

```
// — CONTINUOUS MAIN RUNNER EXECUTION SCRIPT —
public class UnifiedTracker {
    public static void main(String[] args) {
        Scanner inputScanner = new Scanner(System.in);

        System.out.println("=====");
        System.out.println("  WELCOME TO THE GLOBAL STUDENT ENGINE  ");
        System.out.println("=====");

        while (true) {
            System.out.println("\nSelect an action:");
            System.out.println("1. Register a Regular Student");
            System.out.println("2. Register a Graduate Student");
            System.out.println("3. Exit Program");
            System.out.print("Enter choice (1-3): ");
            int menuChoice = inputScanner.nextInt();
            inputScanner.nextLine(); // Clear scanner buffer

            if (menuChoice == 3) {
                System.out.println("\nExiting System... Thank you for using Student Engine. Goodbye!");
                break;
            }

            if (menuChoice != 1 && menuChoice != 2) {
                System.out.println("✗ Invalid option. Please select 1, 2, or 3.");
                continue;
            }

            System.out.print("\nEnter raw student name: ");
            String rawName = inputScanner.nextLine();

            Student currentStudent = null;

            if (menuChoice == 1) {
                currentStudent = new Student(rawName, inputScanner);
            } else if (menuChoice == 2) {
                System.out.print("Enter Research Thesis Topic: ");
                String thesis = inputScanner.nextLine();
                currentStudent = new GraduateStudent(rawName, thesis, inputScanner);
            }
        }
    }
}
```

```
// --- POLYMORPHIC REPORT COMPILATION ---
    System.out.println("\n=====");
    System.out.println("          GENERATING REPORT CARD          ");
    System.out.println("=====");
    System.out.println("Account File:    " + currentStudent.getName());
    System.out.println("Tracked Classes: " + currentStudent.getGrades().length);

    String extraDetails = "";
    if (currentStudent instanceof GraduateStudent) {
        GraduateStudent castedGrad = (GraduateStudent) currentStudent;
        extraDetails = "Research Focus:  " + castedGrad.getThesisTitle() + "\n";
        System.out.print(extraDetails);
    }

    double averageGrade = currentStudent.calculateAverage();
    System.out.println("Final Grade:      " + averageGrade + "%");

    String honorStatus = "Standard Academic Standing";
    if (currentStudent.qualifiesForHonorRoll()) {
        honorStatus = "★ APPROVED FOR HONOR ROLL ★";
    }
    System.out.println("Honor Roll:      " + honorStatus);
    System.out.println("=====");

    // — NEW STEP: AUTOMATICALLY SAVE DATA TO HARD DRIVE VIA FILE I/O —
    // Setting the second parameter to 'true' enables APPEND mode so old entries stay saved!
    try (FileWriter fileWriter = new FileWriter("report_cards.txt", true)) {

        // Write data patterns onto the disk exactly how we formatted them on screen
        fileWriter.write("=== STUDENT RECORD ===\n");
        fileWriter.write("Name: " + currentStudent.getName() + "\n");
        fileWriter.write("Average: " + averageGrade + "%\n");
        if (!extraDetails.isEmpty()) {
            fileWriter.write(extraDetails);
        }
        fileWriter.write("Status: " + honorStatus + "\n");
        fileWriter.write("-----\n\n");

        System.out.println("💾 Backup Status: Report successfully written to 'report_cards.txt'.");

    } catch (IOException e) {
        // Intercept data path anomalies (like read-only file lockouts or hardware drops)
        System.out.println("❌ Local Storage Failure: Could not write record backup to disk drive.");
    }
}

inputScanner.close();
}
```